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Department of Computing and Mathematical Sciences

CO7201: MSc Individual Project

Preliminary Report writing

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https://campus.cs.le.ac.uk/teaching/resources/CO7201/

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Mitigating circumstances

- If a serious event beyond your control has affected your health or personal life and as a result you are unable to submit an assessment on time - you need to let us know
  - This process is known as requesting 'mitigating circumstances'.
- We cannot give extensions directly, all requests for extensions need to go via the Mitigating Circumstances team (cms_mitcircs_support@leicester.ac.uk). This ensures that all students are treated fairly, consistently and confidentially.
- Before submitting a claim, please read the information on these pages carefully:
  - https://le.ac.uk/policies/regulations/mitigating-circumstances

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Project schedule (Spring/Autumn)

Week	Meetings	Class sessions	Deliverable
Pre-	Preliminary meeting	Intro lecture	Project description (P/F)
1		Planning a (SE/Technical) project	
2	Supervision meeting		Preliminary report (Marked)
3	Group Supervision meeting		
4	Supervision meeting		
5		Interim report writing support	
6	Supervision meeting		Interim report (P/F)
7	Principal marker interview		Principal marker interview (P/F)
8	Supervision meeting		
9		Final report writing, and plagiarism in dissertations session	
10	Supervision meeting		Final report template (Formative)
11	Group Supervision meeting		
12	Supervision meeting		
13		Preparing for the viva	Project conclusion (Marked)
14	Viva		



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Project phases (CA7201)

- Topic allocation (7 Feb)
- Project description (14 Feb)
- Start of project (17 Feb) ←
- **Preliminary report (28 Feb)**
- Interim report (28 Mar)
- Interview (31 Mar – 4 Apr)
- Background and Dissertation template (25 Apr)
- Dissertation and project code (16 May)
- Viva (including presentation) (19 – 23 May)



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Checklist ☑

- So far you should have…
  - Read the study guide
  - Contacted your supervisor, finalised a project topic, and set-up fortnightly meetings
  - Set-up Git and be able to commit to your repository
  - Submitted your Project Description via GitLab

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Project Description feedback

- Will be released as it is entered by supervisors
- If it is marked as a ‘fail’, this means you do not currently have a viable project
  - Don’t sit around waiting for “approval”, you need to be working on your Preliminary Report
  - Please take the supervisor comments into consideration when preparing the Preliminary Report. See the study guide for guidance
- Please let me know about any unresponsive supervisors and I can help you to chase them up

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This is **your** project

- An opportunity to show initiative and to take control of your learning
- You are responsible for:
  - finding what you need to know & developing ideas
  - planning and progressing the project
  - delivering on time
- You have to meet your supervisor regularly (once every two weeks) for guidance…
  - …but don't expect them to do the project for you!

Common mistakes

- Waiting too long to start coding, spending too much time doing “research”
  - First two weeks are for planning, you should aim to have a prototype in place for the Principal Marker Interview in Week 7
- Taking too much “inspiration” from existing systems
  - Simply following a few online tutorials is not enough for a 600 hour project on an Advanced MSc course
  - We can only give you credit for the value you add

Project scope (Software Development)

- These are projects where the main contribution is developing a significant software system.
  - Typically, the project requires the student to use their creativity in modelling/developing a non-trivial software system.
  - Although a software product is the main result of projects of this type, the development process (software development tools/techniques, appropriate methodologies/architectures) is relevant.
- Full software development lifecycle for the system:
  - Planning, Requirements, Design, Implementation, Testing, Evaluation



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Project scope (Software Development)

- Many students fail with this type of project by showing insufficient technical challenge/value add. Things to consider:
  - Simple systems are unlikely to provide you with much scope for a high grade. Many existing systems online implement the same functionality already. This is supposed to be an advanced MSc project.
  - Make sure that what you develop is your own work and isn't simply based on existing code or tutorials
  - Add some technically demanding features, that clearly demonstrate your contribution. You should start by identifying a set of requirements, then implement them incrementally (one-by-one), showing your supervisor the work you have done in each meeting.

Project scope (Tech/ML)

- Many students fail with this type of project by showing insufficient technical challenge/value add. Things to consider:
  - interesting, **original** problem with clear criteria for what constitutes a successful solution
  - novel approach or combination of existing algorithms (libraries, tools)
  - iteration of evaluation, feedback and improvement
  - integration into application (e.g. automated data acquisition and presentation in a web app)
  - comparative evaluation demonstrating the performance relative to the state of the art

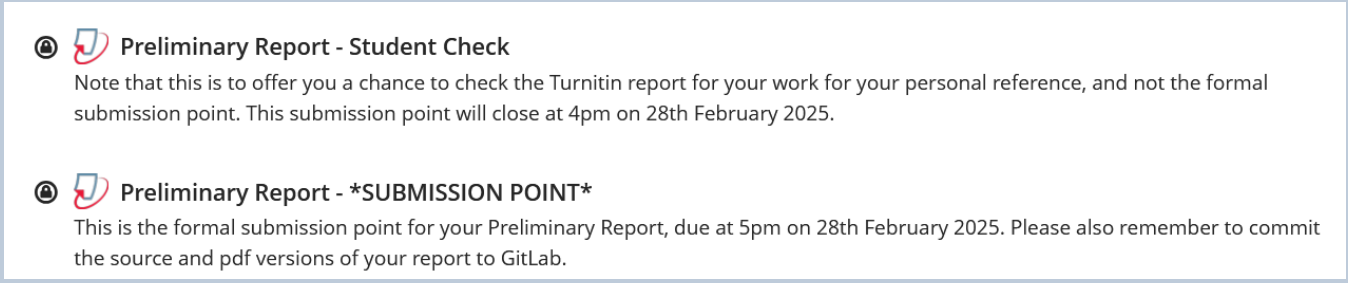
Preliminary Report

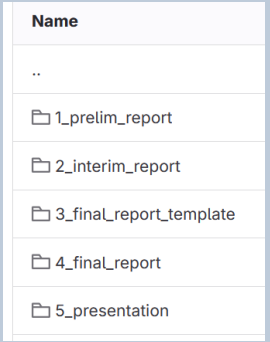
- Scope: 1500 words about 4-6 pages
- Assessment: 5% of final mark
- Deadline: Friday 28th February
- Submission: Blackboard and Gitlab
- Option to check the plagiarism report of your work via Turnitin
  - Will be on Blackboard next week

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Submission (28th Feb)

- Remember to submit your document via Blackboard *and* commit a copy (source and pdf) to your GitLab repository
  - There is a second submission point on Blackboard for you to check the Turnitin report for your work
  - There are separate folders set up automatically for you on GitLab for each document





Preliminary Report

- The preliminary report is a reference for the project's **aims**, **objectives**, **challenges, risks** and the **time plan**.
- You should define/describe these aspects by investigating background material and related work/literature.
- The project's aims, objectives and challenges must be **clearly stated and presented**.
- You should also design viable work and risk plans, to be updated as work progresses.



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Preliminary Report (5%, due 28th Feb)

- A good way to think about this report is as a set of instructions, outlining what needs to be done and how it will be done, in order to deliver the software system that you intend to.
- In theory, you should be able to give this document to ***someone else***, and they would be able to develop this project (or at least have some idea of how to go about it).

The Preliminary Report **must** include:

1. A brief discussion on project relevance/motivations (why is it worth doing it?) and its main challenges
2. A list of requirements (or research questions), including high-level requirements or aims, and detailed requirements or objectives
3. Technical specifications of the project
4. A requirements evaluation plan
5. Background material (including a reading list)
6. A detailed timetable and plan for achieving the objectives of the project, including the milestones of the project and a risk plan.



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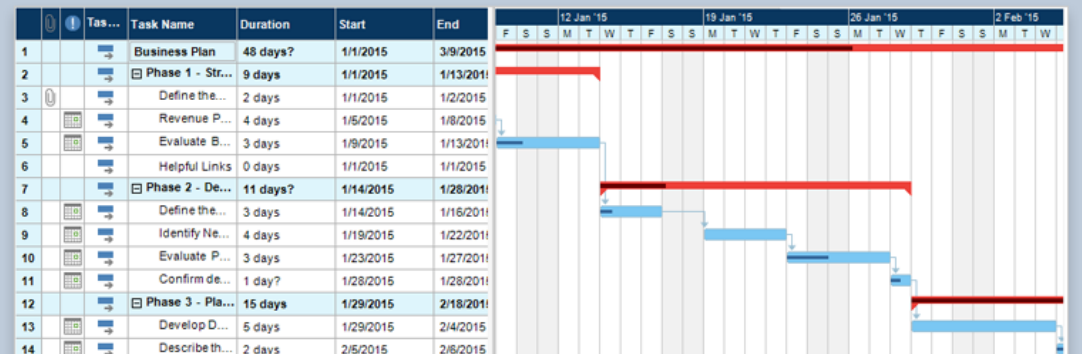
Evaluation of computing projects

- *Good* evaluation is key **to achieving a distinction** in a project
  - What criteria to use to evaluate your system or experimental results?
  - Who will be involved in the evaluation?
  - What type of testing will you do to verify the quality of your software?
- An opportunity to reflect on your performance in your final report

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Plan and timescales

- Detail your project's work plan
- Explain target dates and amount of time required for the completion of aspects of the project
- Be as realistic as you can
  - This will be revisited in the Interim Report



	Tas...	Task Name	Duration	Start	End
1		Business Plan	48 days?	1/1/2015	3/9/2015
2		Phase 1 - Str...	9 days	1/1/2015	1/13/201
3		Define the...	2 days	1/1/2015	1/2/2015
4		Revenue P...	4 days	1/5/2015	1/8/2015
5		Evaluate B...	3 days	1/9/2015	1/13/201
6		Helpful Links	0 days	1/1/2015	1/1/2015
7		Phase 2 - De...	11 days?	1/14/2015	1/28/201
8		Define the...	3 days	1/14/2015	1/16/201
9		Identify Ne...	4 days	1/19/2015	1/22/201
10		Evaluate P...	3 days	1/23/2015	1/27/201
11		Confirm de...	1 day?	1/28/2015	1/28/201
12		Phase 3 - Pla...	15 days	1/29/2015	2/18/201
13		Develop D...	5 days	1/29/2015	2/4/2015
14		Describe th...	2 days	2/5/2015	2/6/2015

Risk plan

- This is asking you to consider how robust the plan you have developed is.
- Estimating the effort required to deliver software is inherently difficult, there is always a risk of (complete or partial) project failure.
- Have you thought about potential pitfalls and things that might go wrong, and how you would go about mitigating that risk?
  - There is no need to provide a detailed plan for each possible failure, but please try to provide alternative strategies (i.e. if A is not possible, we can consider doing B or C).

Things we are looking for (i)

- Is the motivation of the project convincing?
- Do the aims give an informative summary of the project?
- Are the objectives and requirements clearly identified?
  - Are they realistic? Ambitious?

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Things we are looking for (ii)

- Has the relevant background material been identified, well-reviewed and summarized?
  - Has the student found other relevant sources of information on their own?
- Is the plan sufficiently detailed and realistic?
- Has the student set milestones specific to their project?
  - (in addition to the project deliverables)

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Marking rubric (to be circulated)

	Excellent (A)	Good (B)	Pass (C)	Fail (D)
Aims and Objectives (15%)	All project aims and objectives and clearly identified and described, and are challenging at MSc level.	Project aims, objectives and challenges are clearly identified and described, appropriate for advanced MSc level.	Some description of project aims, objectives and challenges although vague and/or incomplete.	Aims, objectives and challenges are extremely vague or missing. It is not clear what will be achieved.
Requirements (20%)	List of requirements is challenging and clearly linked to the aims and objectives, appropriately separated into essential, recommended and optional.	Realistic list of requirements appropriately separated into essential, recommended and optional.	Some requirements are included although they are vague and/or incomplete.	Requirements do not clearly align with the proposed project. It is not clear what will be achieved.
Technical Specification (5%)	In-depth consideration of the tools and techniques required is provided, including details of the languages, platforms, APIs and libraries that will be required to successfully complete the project as appropriate.	Details of appropriate languages, platforms, APIs and libraries that will be required to successfully complete the project are provided.	Some related discussion of tools and techniques is included, but it is not clear how they relate to meeting the aims and objectives.	No technical specification provided.
Requirements Evaluation Plan (10%)	Extensive consideration given to the criteria to be used to evaluate the system/experimental results.	Student has made a meaningful attempt at defining criteria to be used to evaluate the system/experimental results.	Some consideration given to the criteria to be used to evaluate the system/experimental results, but it has been approached in a superficial way.	No meaningful approach to requirements evaluation has been considered.
Background Research and Reading List (10%)	A good deal of relevant background material is identified, well-reviewed and summarized.	Some relevant background material is identified, however, it is not very well connected to the project. It seems more a (reasonable) list of arguments/ literature/ related work that is the result of some reading/searching activity.	The background material presented is not really relevant and not reviewed or summarized.	No or little background material/literature was identified or researched.
Time-plan and Risk Plan (20%)	An excellent, feasible, well-written plan describes how the project objectives will be achieved. Appropriate tools (such as Gantt charts) used to convey the plan.	Overall, the student did a good job defining the project and describing their plan for doing it. However, the plan could still be improved.	The plan is reasonable, but requires refinement to ensure it can be realistically followed.	There is no project plan at all or there is a vague or ill-thought plan. It is not clear what the student will do.
Overall Impression (20%)	Overall, the student did an excellent job defining the project and describing their plan for completing it.	Overall, the work is of good quality. However, it could have been better organised and/or provided deeper analysis of the core aspects of the project	One can understand what the project is about and roughly how the student will work on it, although lots of details are missing and the general impression is that the report was not well thought out and prepared.	No submission or little to no attempt to address the task. Virtually no mention of anything related to the project, related work or references/information resources. No planning and no time-scales.



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Plagiarism and GitLab

- I have to deal with plagiarism issues in the project module every year
  - When writing, you shouldn't ever need to be using the copy/paste functions
- Good use of GitLab can protect you from accusations of plagiarism
  - You should be committing multiple times per week, if not multiple times per day



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Plagiarism: what happens if you are caught?

- Typically your project will be marked 0.
- If you fail the project, you cannot get an MSc (you may get a postgraduate diploma instead).
- Your transcript will record that you scored 0 in the project. Employers will want to know why.
- If an employer asks us whether you are honest, we are legally obliged to tell them that you plagiarized.

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Further resources

- Read study guide.
- Book: Projects in Computing and Information Systems. A student's guide. 2nd ed. Christian W. Dawson
- More information on the web page:

https://campus.cs.le.ac.uk/teaching/resources/CO7201/

  - Calendar
  - Information about plagiarism
  - Student Support and Development Service
  - Presentation and writing skills
  - Templates for deliverables

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Questions?

- I have allocated the rest of the session to answering your questions
- I will circulate a copy of these slides, the Preliminary Report template, and the rubric via email following this session